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Inventor(s)	Hong; Suk Joon

SECONDARY BATTERY

Abstract

Disclosed is a secondary battery capable of reducing and/or eliminating an external terminal structure, thereby reducing the number of parts and production costs thereof. The secondary battery includes an electrode assembly including a first electrode plate and a second electrode plate, a first case mounted in a shape surrounding at least one surface of the electrode assembly and electrically connected to the first electrode plate, a second case mounted in a shape surrounding at least the opposite surface of the electrode assembly and electrically connected to the second electrode plate, and an insulating gasket located between the first case and the second case and made of an insulative material to block electrical connection between the first case and the second case.

Inventors:	Hong; Suk Joon (Yongin-si, KR)
Applicant:	Samsung SDI Co., Ltd. (Yongin-si, KR)
Family ID:	1000008180412
Assignee:	Samsung SDI Co., Ltd. (Yongin-si, KR)
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Background/Summary

CROSS-REFERENCE TO THE RELATED APPLICATIONS [0001] This present application claims priority to and the benefit under 35 U.S.C § 119(a)-(d) of Korean Patent Application No. 10-2024-0033158, filed on Mar. 8, 2024, in the Korean Intellectual Property Office, the entire disclosure of which is incorporated herein by reference.

FIELD

[0002] The present disclosure relates to a secondary battery.

BACKGROUND

[0003] Unlike a primary battery that cannot be recharged, a secondary battery is a battery that can be recharged and discharged. A low-capacity secondary battery may be used for portable small-sized electronic devices, such as smartphones, feature phones, notebook computers, digital cameras, and camcorders, and a high-capacity secondary battery may be used as a power source for driving a motor and a power storage battery in hybrid vehicles or electric vehicles. The secondary battery may include an electrode assembly having a positive electrode and a negative electrode, a case accommodating the electrode assembly, an electrode terminal connected to the electrode assembly, and the like.

[0004] The above-described information disclosed in the technology that serves as the background of the present disclosure is only for improving understanding of the background of the present disclosure and thus may include information that does not constitute the related art.

SUMMARY

[0005] The present disclosure provides a secondary battery capable of reducing and/or eliminating an external terminal structure, thereby reducing the number of parts and production costs thereof.

[0006] These and other aspects and features of the present disclosure will be described in or will be apparent from the following description of embodiments of the present disclosure.

[0007] In order to accomplish the above objective, an exemplary secondary battery according to some embodiments of the present disclosure includes an electrode assembly including a first electrode plate and a second electrode plate, a first case mounted in a shape surrounding at least one surface of the electrode assembly and electrically connected to the first electrode plate, a second case mounted in a shape surrounding at least an opposite surface of the electrode assembly and electrically connected to the second electrode plate, and an insulating gasket located between the first case and the second case, the insulating gasket comprising an insulative material, wherein electrical connection between the first case and the second case is blocked.

[0008] In some embodiments, the secondary battery may further include a first connection tab connected to the first electrode plate and extending above the electrode assembly and/or in a direction towards the first case, such that the first connection tab is electrically connected to the first case.

[0009] In some embodiments, the secondary battery may further include a second connection tab connected to the second electrode plate and extending below the electrode assembly and/or in a direction towards the second case such that the second connection tab is electrically connected to the second case.

[0010] In some embodiments, the first connection tab may be connected to one side of the first electrode plate in the longitudinal direction of the first electrode plate.

[0011] In some embodiments, the second connection tab may be connected to one side of the second electrode plate in the longitudinal direction of the second electrode plate.

[0012] In some embodiments, the first connection tab may be welded to the first case, and the second connection tab may be welded to the second case.

[0013] In some embodiments, the second case may be disposed on the first case, and an first end portion of the second case may be located inside the first case.

[0014] In some embodiments, the insulating gasket may be mounted in a shape surrounding the first end portion of the second case to block electrical connection between the first case and the second case.

[0015] In some embodiments, the insulating gasket may include an inner gasket abutting an inner surface of the second case, an outer gasket abutting an outer surface of the second case, and a connection gasket interconnecting the inner gasket and the outer gasket.

[0016] In some embodiments, the length of the outer gasket in a vertical direction may be greater than the length of the inner gasket in the vertical direction.

[0017] In some embodiments, the length of the second case in the vertical direction may be greater than the length of the outer gasket in the vertical direction.

[0018] In some embodiments, the length of the outer gasket in the vertical direction may be greater than the length of the first case in the vertical direction.

[0019] In order to accomplish the above objective, an exemplary secondary battery according to some embodiments of the present disclosure includes an electrode assembly including a first electrode plate and a second electrode plate, a first case configured to cover a first side of the electrode assembly and electrically connected to the first electrode plate, a second case electrically connected to the second electrode plate and configured to accommodate the electrode assembly, the second case including a side covered by the first case, and an insulating gasket located between the first case and the second case, the insulating gasket comprising an insulative material.

[0020] In some embodiments, the first case and the second case may be formed as conductors and may have different polarities.

[0021] In some embodiments, the insulating gasket may include an inner gasket abutting an inner surface of the second case, an outer gasket abutting an outer surface of the second case, and a connection gasket interconnecting the inner gasket and the outer gasket.

[0022] In some embodiments, the first case may include a first case body formed to allow a first connection tab connected to the first electrode plate to be electrically connected thereto and to cover the first side of the electrode assembly and a first extension member extending away from the edge of the first case body to be located outside the outer gasket.

[0023] In some embodiments, the second case may include a second case body formed to allow a second connection tab connected to the second electrode plate to be electrically connected thereto and to cover a second side of the electrode assembly and a second extension member extending away from the edge of the second case body to be located between the outer gasket and the inner gasket.

[0024] In some embodiments, the length of the second extension member in a vertical direction may be greater than the length of the outer gasket in the vertical direction.

[0025] In some embodiments, the length of the outer gasket in the vertical direction may be greater than the length of the first extension member in the vertical direction.

[0026] In some aspects, methods of manufacturing a secondary battery are generally described. In some embodiments, the method comprises: providing an insulating gasket located between a first case and a second case such that electrical connection between the first case and the second case is blocked, wherein: the first case mounted in a shape surrounding at least one surface of an electrode assembly, the electrode assembly comprising a first electrode plate and a second electrode plate, and electrically connected to the first electrode plate; the second case mounted in a shape surrounding at least an opposite surface of the electrode assembly and electrically connected to the second electrode plate; and the insulating gasket comprises an insulative material.

[0027] In some embodiments, the insulating gasket is mounted in a shape surrounding a first end

portion of the second case to block electrical connection between the first case and the second case. [0028] In some embodiments, a first connection tab, the first connection tab connected to the first electrode plate and extending away from the electrode assembly and/or in a direction towards the first case, electrically connects the first case to the first electrode plate; and a second connection tab, the second connection tab connected to the second electrode plate and extending below the electrode assembly and/or in a direction towards the second case, electrically connects the second case to the second electrode plate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0029] The following drawings attached to the present specification illustrate embodiments of the present disclosure, and further describe aspects and features of the present disclosure together with the detailed description of the present disclosure. Thus, the present disclosure should not be construed as being limited to the drawings:

[0030] FIGS. 1 and 2 are, respectively, a perspective view and a cross-sectional view showing an exemplary secondary battery according to some embodiments of the present disclosure;

[0031] FIG. 3 is a cross-sectional view showing the configuration of the exemplary secondary battery according to some embodiments of the present disclosure in an exploded manner;

[0032] FIG. 4 is an exploded perspective view of the exemplary secondary battery according to some embodiments of the present disclosure;

[0033] FIG. 5 is a cross-sectional view of an exemplary electrode assembly according to some embodiments of the present disclosure;

[0034] FIG. 6 is a perspective view showing an exemplary insulating gasket according to some embodiments of the present disclosure;

[0035] FIG. 7 is a cross-sectional view showing the exemplary insulating gasket according to some embodiments of the present disclosure;

[0036] FIG. 8 is a perspective view showing an exemplary secondary battery according to some embodiments of the present disclosure;

[0037] FIG. 9 is an exploded perspective view of the exemplary secondary battery according to some embodiments of the present disclosure;

[0038] FIGS. 10A and 10B are perspective views showing a battery pack including the exemplary secondary battery according to some embodiments of the present disclosure; and

[0039] FIGS. 11A and 11B are, respectively, a perspective view and a side view showing vehicles including the exemplary battery pack according to some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0040] Hereinafter, embodiments of the present disclosure will be described, in detail, with reference to the accompanying drawings. The terms or words used in the present specification and claims are not to be limitedly interpreted as general or dictionary meanings and should be interpreted as meanings and concepts that are consistent with the technical idea of the present disclosure on the basis of the principle that an inventor can be his/her own lexicographer to appropriately define concepts of terms to describe his/her invention in the best way.

[0041] The embodiments described in this specification and the configurations shown in the drawings are only some of the embodiments of the present disclosure and do not represent all of the technical spirit, aspects, and features of the present disclosure. Accordingly, it should be understood that there may be various equivalents and modifications that can replace or modify the embodiments described herein at the time of filing this application.

[0042] In addition, the terms “comprise” or “include” and/or “comprising” or “including,” when used in this specification, specify the presence of stated shapes, numbers, steps, operations,

members, elements, and/or groups thereof, but do not preclude the presence or addition of one or more other shapes, numbers, steps, operations, members, elements, and/or groups thereof.

[0043] Additionally, for the purpose of facilitating an understanding of the invention, the attached drawings are not depicted to actual scale; dimensions of some components may be exaggerated for clarity. Also, identical components in different embodiments may be denoted with the same reference numerals.

[0044] When two objects of comparison are referred to as being the same, it means the two objects are “substantially the same.” Thus, substantially the same may include a deviation that is considered low in the art, for example, a deviation of less than 5%. In addition, when a parameter is said to be uniform in a certain region, it may mean that the parameter is uniform from an average perspective.

[0045] Although “first,” “second,” and the like are used to describe various components, these components are of course not limited by these terms. These terms are only used to distinguish one component from other components, and unless otherwise stated, a first component could be termed a second component.

[0046] Throughout the specification, unless specifically stated otherwise, each component may be singular or plural.

[0047] When an arbitrary element is referred to as being “disposed above (or below)” or “disposed on (or under)” a component, it may mean not only that the arbitrary element is disposed in contact with an upper surface (or lower surface) of the component, but also that other elements may be interposed between the component and the arbitrary element disposed on (or under) the component.

[0048] When a component is described as being “connected,” “coupled,” or “joined” to another component within this patent document, it is understood that the components may be directly connected or joined to each other. However, it should also be interpreted that an intervening component may be interposed between them, or that each component may be “connected,” “coupled,” or “joined” through another intermediary component. Furthermore, when one part is described as being electrically connected (electrically coupled) to another, this encompasses not only a direct connection but also includes scenarios where other elements are positioned in between, facilitating an indirect connection.

[0049] Throughout this specification, the term ‘A and/or B’ should be interpreted as meaning either A, B, or both A and B, unless an alternative interpretation is explicitly stated. Thus, ‘and/or’ encompasses any and all possible combinations of the items listed. Similarly, when ‘C to D’ is mentioned, it is understood to mean C or more, up to and including D, unless noted otherwise. The terminology employed herein is intended solely for describing specific embodiments and should not be regarded as limiting the scope of this disclosure.

[0050] FIGS. **1** and **2** are, respectively, a perspective view and a cross-sectional view showing an exemplary secondary battery **100** according to some embodiments of the present disclosure. FIG. **2** is a cross-sectional view taken along line 2-2 in FIG. **1**. As shown in FIGS. **1** and **2**, the exemplary secondary battery **100** according to some embodiments of the present disclosure may include an electrode assembly **120**, a first case **150**, a second case **160**, and an insulating gasket **170**. In some examples, the exemplary secondary battery **100** according to the present disclosure may further include at least one of a first connection tab **130** or a second connection tab **140**. In the present disclosure, the secondary battery **100** may be referred to as a battery.

[0051] The first case **150** and the second case **160** may be mounted with the insulating gasket **170** interposed therebetween, and may serve to accommodate the electrode assembly **120** and an electrolyte and/or to define the external appearance of the secondary battery **100**. Each of the first case **150** and the second case **160** may include or be referred to as a can, a housing, or an exterior body. The first case **150** and the second case **160** may include metal, such as steel, stainless steel (SUS), nickel-plated steel, a steel alloy, aluminum, an aluminum alloy, and/or deep-drawing cold-rolled steel (SPCE), and/or a laminated film or plastic comprising a pouch. At least one of the first

case **150** or the second case **160** may be provided with a beading portion (not shown) that is depressed inwardly. The beading portion may protrude toward the interior of the first case **150** and the second case **160** to suppress movement of the electrode assembly **120**.

[0052] The electrode assembly **120** may be accommodated together with an electrolyte in a space defined by the first case **150** and the second case **160**. The electrode assembly **120** may include or be referred to as an electrode group, an electrode body, or a jellyroll. The electrode assembly **120** may include a first electrode plate **121**, a second electrode plate **122**, and a separator **123** (as shown in FIG. 5) interposed between the first electrode plate **121** and the second electrode plate **122**. The components of the electrode assembly **120** may be stacked in a plate shape. The electrode assembly **120** may be formed in various configurations, for example, a stacked configuration or a cylindrically wound configuration. The electrode assembly **120** may be configured such that the first electrode plate **121**, the separator **123**, and the second electrode plate **122** are stacked in that order. In some embodiments, the electrode assembly **120** may be configured such that the first electrode plate **121**, the second electrode plate **122**, and the separator **123**, each of which has a plate shape including a small-width portion and a large-width portion consecutively connected to each other, are stacked.

[0053] FIG. 3 is a cross-sectional view showing the configuration of the exemplary secondary battery **100** according to some embodiments of the present disclosure in an exploded manner, FIG. 4 is an exploded perspective view of the exemplary secondary battery **100** according to some embodiments of the present disclosure, and FIG. 5 is a cross-sectional view of the exemplary electrode assembly **120** according to some embodiments of the present disclosure. As shown in FIGS. 3 to 5, the first electrode plate **121** may include a first substrate **1211** and a first active material layer **1212** located on the first substrate **1211**. The first substrate **1211** may include a first uncoated portion or a first tab **1213** on which the first active material layer **1212** is not located. The first uncoated portion or the first tab **1213** of the first substrate **1211** may extend outward (e.g., in a lateral direction), and the first tab **1213** may be electrically connected to the first connection tab **130**. In some embodiments of the present disclosure, the first tab **1213** may be referred to as a first uncoated portion or a positive electrode substrate tab.

[0054] The second electrode plate **122** may include a second substrate **1221** and a second active material layer **1222** located on the second substrate **1221**. The second substrate **1221** may include a second uncoated portion or a second tab **1223** on which the second active material layer **1222** is not located. The second uncoated portion or the second tab **1223** of the second substrate **1221** may extend outward (e.g., in a lateral direction), and the second tab **1223** may be electrically connected to the second connection tab **140**. In some embodiments, the first tab **1213** and the second tab **1223** may be implemented in various forms, for example, may extend in the same direction or may extend in different directions. In some embodiments of the present disclosure, the second tab **1223** may be referred to as a second uncoated portion or a negative electrode substrate tab.

[0055] The first electrode plate **121** may function as a positive electrode. In this case, the first substrate **1211** may be formed as, for example, an aluminum foil, and the first active material layer **1212** may include, for example, a transition metal oxide. The second electrode plate **122** may function as a negative electrode. In this case, the second substrate **1221** may be formed as, for example, a copper foil and/or a nickel foil, and the second active material layer **1222** may include, for example, graphite and/or silicone.

[0056] The separator **123** may prevent short circuit between the first electrode plate **121** and the second electrode plate **122** while allowing lithium ions to move therebetween. In some embodiments, the separator **123** may be located on each of two opposite side surfaces of the first electrode plate **121** or may be located on each of two opposite side surfaces of the second electrode plate **122**.

[0057] As the positive electrode active material, a compound capable of reversibly intercalating/deintercalating lithium (e.g., a lithiated intercalation compound) may be used. For

example, at least one of a composite oxide of lithium and a metal selected from cobalt, manganese, nickel, and combinations thereof may be used.

[0058] The composite oxide may be a lithium transition metal composite oxide, and examples thereof may include a lithium nickel-based oxide, a lithium cobalt-based oxide, a lithium manganese-based oxide, a lithium iron phosphate-based compound, a cobalt-free nickel-manganese-based oxide, or a combination thereof.

[0059] As an example, a compound represented by any one of the following formulas may be used:

$\text{Li}_{1-a}\text{A}_{a-1-b}\text{X}_b\text{O}_{2-c}\text{D}_c$ ($0.90 \leq a \leq 1.8$, $0 \leq b \leq 0.5$, $0 \leq c \leq 0.05$);

$\text{Li}_{1-a}\text{Mn}_{a-2-b}\text{X}_b\text{O}_{4-c}\text{D}_c$ ($0.90 \leq a \leq 1.8$, $0 \leq b \leq 0.5$, $0 \leq c \leq 0.05$);

$\text{Li}_{1-a}\text{Ni}_{a-1-b-c}\text{Co}_b\text{X}_c\text{O}_{2-\alpha}\text{D}_\alpha$ ($0.90 \leq a \leq 1.8$, $0 \leq b \leq 0.5$, $0 \leq c \leq 0.5$, $0 < \alpha < 2$);

$\text{Li}_{1-a}\text{Ni}_{a-1-b-c}\text{Mn}_b\text{X}_c\text{O}_{2-\alpha}\text{D}_\alpha$ ($0.90 \leq a \leq 1.8$, $0 \leq b \leq 0.5$, $0 \leq c \leq 0.5$, $0 < \alpha < 2$);

$\text{Li}_{1-a}\text{Ni}_b\text{Co}_c\text{L}_1\text{G}_d\text{O}_{2e}$ ($0.90 \leq a \leq 1.8$, $0 \leq b \leq 0.9$, $0 \leq c \leq 0.5$, $0 \leq d \leq 0.5$, $0 \leq e \leq 0.1$); $\text{Li}_{1-a}\text{Ni}_b\text{G}_b\text{O}_{2e}$ ($0.90 \leq a \leq 1.8$, $0.001 \leq b \leq 0.1$);

$\text{Li}_{1-a}\text{Co}_b\text{G}_b\text{O}_{2e}$ ($0.90 \leq a \leq 1.8$, $0.001 \leq b \leq 0.1$);

$\text{Li}_{1-a}\text{Mn}_{a-1-b}\text{G}_b\text{O}_{2e}$ ($0.90 \leq a \leq 1.8$, $0.001 \leq b \leq 0.1$);

$\text{Li}_{1-a}\text{Mn}_{a-2}\text{G}_b\text{O}_{4e}$ ($0.90 \leq a \leq 1.8$, $0.001 \leq b \leq 0.1$);

$\text{Li}_{1-a}\text{Mn}_{a-1-g}\text{G}_g\text{PO}_4$ ($0.90 \leq a \leq 1.8$, $0 \leq g \leq 0.5$); $\text{Li}_{1-f}\text{Fe}_2(\text{PO}_4)_3$ ($0 \leq f \leq 2$); Li_aFePO_4 ($0.90 \leq a \leq 1.8$).

[0060] In the above formulas: A is Ni, Co, Mn, or a combination thereof; X is Al, Ni, Co, Mn, Cr, Fe, Mg, Sr, V, a rare earth element, or a combination thereof; D is O, F, S, P, or a combination thereof; G is Al, Cr, Mn, Fe, Mg, La, Ce, Sr, V, or a combination thereof; and L1 is Mn, Al, or a combination thereof.

[0061] A positive electrode for a lithium secondary battery may include a current collector and a positive electrode active material layer formed on the current collector. The positive electrode active material layer may include a positive electrode active material and may further include a binder and/or a conductive material.

[0062] The content of the positive electrode active material is in a range of about 90 wt % to about 99.5 wt % on the basis of 100 wt % of the positive electrode active material layer, and the content of the binder and the conductive material is in a range of about 0.5 wt % to about 5 wt %, respectively, on the basis of 100 wt % of the positive electrode active material layer.

[0063] The current collector may be aluminum (Al) but is not limited thereto.

[0064] The negative electrode active material may include a material capable of reversibly intercalating/deintercalating lithium ions, lithium metal, an alloy of lithium metal, a material capable of being doped and undoped with lithium, or a transition metal oxide.

[0065] The material capable of reversibly intercalating/deintercalating lithium ions may be a carbon-based negative electrode active material, which may include, for example, crystalline carbon, amorphous carbon, or a combination thereof. Examples of the crystalline carbon may include graphite, such as natural graphite or artificial graphite, and examples of the amorphous carbon may include soft carbon, hard carbon, a pitch carbide, a meso-phase pitch carbide, sintered coke, and the like.

[0066] A Si-based negative electrode active material or a Sn-based negative electrode active material may be used as the material capable of being doped and undoped with lithium. The Si-based negative electrode active material may be silicon, a silicon-carbon composite, SiO_x ($0 < x < 2$), a Si-based alloy, or a combination thereof.

[0067] The silicon-carbon composite may be a composite of silicon and amorphous carbon. According to one embodiment, the silicon-carbon composite may be in the form of a silicon particle and amorphous carbon coated on the surface of the silicon particle.

[0068] The silicon-carbon composite may further include crystalline carbon. For example, the silicon-carbon composite may include a core including crystalline carbon and silicon particle and an amorphous carbon coating layer on the surface of the core.

[0069] A negative electrode for a lithium secondary battery may include a current collector and a negative electrode active material layer disposed on the current collector. The negative electrode active material layer may include a negative electrode active material and may further include a binder and/or a conductive material.

[0070] For example, the negative electrode active material layer may include about 90 wt % to about 99 wt % of a negative electrode active material, about 0.5 wt % to about 5 wt % of a binder, and about 0 wt % to about 5 wt % of a conductive material.

[0071] A non-aqueous binder, an aqueous binder, a dry binder, or a combination thereof may be used as the binder. When an aqueous binder is used as the negative electrode binder, a cellulose-based compound capable of imparting viscosity may be further included.

[0072] As the negative electrode current collector, one selected from copper foil, nickel foil, stainless steel foil, titanium foil, nickel foam, copper foam, conductive metal-coated polymer substrate, and combinations thereof may be used.

[0073] An electrolyte for a lithium secondary battery may include a non-aqueous organic solvent and a lithium salt.

[0074] The non-aqueous organic solvent acts as a medium through which ions involved in the electrochemical reaction of the battery can move.

[0075] The non-aqueous organic solvent may be a carbonate-based, an ester-based, an ether-based, a ketone-based, an alcohol-based solvent, an aprotic solvent, and may be used alone or in combination of two or more.

[0076] In addition, when a carbonate-based solvent is used, a mixture of cyclic carbonate and chain carbonate may be used.

[0077] Depending on the type of lithium secondary battery, a separator may be present between the first electrode plate (e.g., the negative electrode) and the second electrode plate (e.g., the positive electrode). As the separator, polyethylene, polypropylene, polyvinylidene fluoride, or a multilayer film of two or more layers thereof may be used.

[0078] The separator may include a porous substrate and a coating layer including an organic material, an inorganic material, or a combination thereof on one or both surfaces of the porous substrate.

[0079] The organic material may include a polyvinylidene fluoride-based heavy antibody or a (meth)acrylic polymer.

[0080] The inorganic material may include inorganic particles selected from Al_2O_3 , SiO_2 , TiO_2 , SnO_2 , CeO_2 , MgO , NiO , CaO , GaO , ZnO , ZrO_2 , Y_2O_3 , SrTiO_3 , BaTiO_3 , $\text{Mg}(\text{OH})_2$ boehmite, and combinations thereof but is not limited thereto.

[0081] The organic material and the inorganic material may be mixed in one coating layer or may be in the form of a coating layer containing an organic material and a coating layer containing an inorganic material that are laminated on each other.

[0082] The first connection tab **130** may be connected to the first electrode plate **121**, and may extend above the electrode assembly **120** and/or in a direction towards the first case such that the first connection tab **130** is electrically connected to the first case **150**. The first connection tab **130** may be electrically connected to the first tab **1213** provided at the first electrode plate **121**. One side of the first connection tab **130** may be electrically connected to the first electrode plate **121** of the electrode assembly **120**, and the other side of the first connection tab **130** may be fixed to the inner surface of the first case **150**. The first connection tab **130** may be fixed to the first case **150** through welding. The first connection tab **130** may be connected to one side (left side in FIG. 4) of the first electrode plate **121** in the longitudinal direction (e.g., in a direction parallel to the maximum dimension of the first electrode plate **121**) of the first electrode plate **121**. In some embodiments, the first connection tab **130** may be a strip-shaped conductor. A plurality of first tabs **1213** extending and/or protruding from the electrode assembly **120** may be electrically connected

to one side of the first connection tab **130**.

[0083] The first connection tab **130** may include copper, a copper alloy, nickel, a nickel alloy, aluminum, and/or an aluminum alloy. In some embodiments, the first connection tab **130** may be bent above the electrode assembly **120** to be fixed to the inner side of a first case body **152** through welding. Thus, the first electrode plate **121** and the first case **150** may have the same polarity.

[0084] The second connection tab **140** may be connected to the second electrode plate **122**, and may extend below the electrode assembly **120** and/or in a direction towards the second case such that the second connection tab **140** is electrically connected to the second case **160**. The second connection tab **140** may be electrically connected to the second tab **1223** provided at the second electrode plate **122**. One side of the second connection tab **140** may be electrically connected to the second electrode plate **122** of the electrode assembly **120**, and the other side of the second connection tab **140** may be fixed to the inner surface of the second case **160**. The second connection tab **140** may be fixed to the second case **160** through welding. The second connection tab **140** may be connected to one side (left side in FIG. 4) of the second electrode plate **122** in the longitudinal direction of the second electrode plate **122**. In some embodiments, the second connection tab **140** may be a strip-shaped conductor. A plurality of second tabs **1223** extending and/or protruding from the electrode assembly **120** may be electrically connected to one side of the second connection tab **140**.

[0085] The second connection tab **140** may include copper, a copper alloy, nickel, a nickel alloy, aluminum, and/or an aluminum alloy. In some embodiments, the second connection tab **140** may be bent below the electrode assembly **120** to be fixed to the inner side of a second case body **162** through welding. Thus, the second electrode plate **122** and the second case **160** may have the same polarity.

[0086] The first case **150** may be variously modified, so long as the first case **150** is mounted in a shape surrounding one surface of the electrode assembly **120** and is electrically connected to the first electrode plate **121**. In some embodiments, the first case **150** may cover the upper side of the electrode assembly **120** and may be electrically connected to the first electrode plate **121**. The first case **150** according to an embodiment of the present disclosure may include a first case body **152** and a first extension member **154**. The first case body **152** may be variously modified, so long as the first connection tab **130** connected to the first electrode plate **121** is electrically connected to the first case body **152**, and the first case body **152** covers the upper side of the electrode assembly **120**. The first extension member **154** may extend downward from the edge of the first case body **152** (e.g., away from the edge of the first case body **152** and/or towards the electrode assembly **120**) and may be located outside an outer gasket **174**. The shape of the first case **150** and the shape of the second case **160** may correspond to each other. The first case **150** may include a small-width portion and a large-width portion, which are connected to each other to form one part. The first case body **152** may have a plate shape. The width of one side of the first case body **152** in the longitudinal direction thereof may be smaller than the width of the opposite side of the first case body **152** in the longitudinal direction thereof. The first extension member **154** may be formed along the edge of the first case body **152** (e.g., along a perimeter of the first case body **152**) and may extend downward therefrom (e.g., away from the edge of the first case body **152** and/or towards the electrode assembly **120**). The first case **150** may have an open lower side and may be mounted so as to surround the upper side of the electrode assembly **120**. The first case **150** may be mounted in contact with the outer surface of the insulating gasket **170**. The first case **150** may include at least one of aluminum, steel, stainless steel, injection-molded plastic, a polymer composite material, and/or glass fiber reinforced plastic.

[0087] The second case **160** may be variously modified, so long as the second case **160** is mounted in a shape surrounding the opposite surface of the electrode assembly **120** and is electrically connected to the second electrode plate **122**. In some embodiments, the second case **160** may cover the lower side of the electrode assembly **120** and may be electrically connected to the second

electrode plate **122**. The second case **160** may be variously modified, so long as the second case **160** accommodates the electrode assembly **120** and the upper side thereof is covered by the first case **150**. The second case **160** may be located beneath the first case **150** and/or disposed on the first case **150**, and the upper end portion of the second case **160** may be located inside the first case **150**. The first case **150** and the second case **160** may be formed as conductors and may have different polarities.

[0088] The second case **160** according to some embodiments of the present disclosure may include a second case body **162** and a second extension member **164**. The second case body **162** may be variously modified, so long as the second connection tab **140** connected to the second electrode plate **122** is electrically connected to the second case body **162**, and the second case body **162** covers the lower side of the electrode assembly **120**. The second extension member **164** may extend upward from the edge of the second case body **162** and may be located between the outer gasket **174** and an inner gasket **172**.

[0089] The second case **160** may include a small-width portion and a large-width portion, which are connected to each other to form one part. The second case body **162** may have a plate shape. The width of one side of the second case body **162** in the longitudinal direction thereof may be smaller than the width of the opposite side of the second case body **162** in the longitudinal direction thereof. The second extension member **164** may be formed along the edge of the second case body **162** and may extend upward therefrom. The second case **160** may have an open upper side and may be mounted so as to surround the lower side of the electrode assembly **120**. The second extension member **164** of the second case **160** may be mounted so as to be inserted into the insulating gasket **170**. The second case **160** may include at least one of aluminum, steel, stainless steel, injection-molded plastic, a polymer composite material, and/or glass fiber reinforced plastic.

[0090] FIG. **6** is a perspective view showing the exemplary insulating gasket **170** according to the present disclosure, and FIG. **7** is a cross-sectional view showing the exemplary insulating gasket **170** according to the present disclosure. As shown in FIGS. **4**, **6**, and **7**, the insulating gasket **170** may be mounted in an interference fit manner between the first case **150** and the second case **160**, which are conductors, thereby blocking electrical connection between the first case **150** and the second case **160**. The first connection tab **130** and the second connection tab **140** may be connected to the first case **150** and the second case **160**, respectively, which are electrically disconnected from each other. The insulating gasket **170** may be variously modified, so long as the insulating gasket **170** is located between the first case **150** and the second case **160** and is made of an insulative material to block electrical connection between the first case **150** and the second case **160**. In some embodiments, the insulating gasket **170** may be mounted in a shape surrounding the upper end portion of the second case **160**, thereby blocking electrical connection between the first case **150** and the second case **160**. Because the insulating gasket **170** is mounted so as to surround the upper end portion of the second case **160** and the first case **150** is mounted outside the insulating gasket **170**, it may be possible to prevent the occurrence of short circuit between the first case **150** and the second case **160**. In some embodiments, the insulating gasket **170** may include an inner gasket **172** abutting the inner surface of the second case **160**, an outer gasket **174** abutting the outer surface of the second case **160**, and a connection gasket **176** interconnecting the inner gasket **172** and the outer gasket **174**.

[0091] The insulating gasket **170** may include an insulative material to block electrical connection. The insulating gasket **170** may use polypropylene (PP) as a base material and may use polybutylene terephthalate (PBT) for heat resistance. In some embodiments, the insulating gasket **170** may include polypropylene, polyethylene, and/or ethylene propylene diene terpolymer (EPDM), none of which reacts with the electrolyte. Further, the insulating gasket **170** may include at least one of polyimide, Teflon, and/or polyester.

[0092] Polyimide has excellent heat resistance and electrical insulation performance and is chemically stable, so the same exhibits excellent insulation performance in the chemical

environment in a battery. Teflon has relatively high heat resistance, is chemically stable, and has excellent electrical insulation performance. Polyethylene is relatively flexible and has good electrical insulation performance and durability. Polyester has excellent durability and is electrically and chemically stable. Polypropylene is an economical and durable material. EPDM is a thermally and chemically stable rubber that is mainly used as an insulating and sealing material. [0093] The outer gasket **174** and the inner gasket **172** may extend in the vertical direction. The upper end of the outer gasket **174** and the upper end of the inner gasket **172** may face each other and may be connected to each other via the connection gasket **176**. The insulating gasket **170** may have an inverted U-shaped longitudinal-section (e.g., a U-shaped cross-section) having an open lower side. The insulating gasket **170** may be elastic, and may be mounted so as to surround the upper end portion of the second case **160**. In some embodiments, the length A of the outer gasket **174** in the vertical direction may be greater than the length B (as shown in FIG. 3) of the inner gasket **172** in the vertical direction. The outer gasket **174** may prevent contact between the first extension member **154** of the first case **150** and the second extension member **164** of the second case **160**. In some embodiments, the length C of the second case **160** in the vertical direction may be greater than the length A of the outer gasket **174** in the vertical direction. This may mean that the length of the second extension member **164** in the vertical direction is greater than the length A of the outer gasket **174** in the vertical direction. Thus, it may be possible to prevent the lower end of the outer gasket **174** from protruding downward below the second case **160**. In some embodiments, the length A of the outer gasket **174** in the vertical direction may be greater than the length D of the first case **150** in the vertical direction. This may mean that the length A of the outer gasket **174** in the vertical direction is greater than the length of the first extension member **154** in the vertical direction. If the length D of the first case **150** in the vertical direction is greater than the length A of the outer gasket **174** in the vertical direction, the first case **150** may face the second case **160**, and a short circuit may occur therebetween. In some embodiments, it may be possible to ensure electrical safety by setting the length A of the outer gasket **174** in the vertical direction to be greater than the length D of the first case **150** in the vertical direction.

[0094] The insulating gasket **170** may be compressed by movement of the first case **150**, and thus the gap between the first case **150** and the second case **160** may be shielded, whereby leakage of the electrolyte may be prevented.

[0095] Because the upper end portion of the second extension member **164** provided at the second case **160** is inserted into the space defined between the inner gasket **172** and the outer gasket **174**, the insulating gasket **170** may be secured to the second case **160**. Electrical connection between the second connection tab **140** and the second case **160** may be achieved in a state in which the electrode assembly **120** is accommodated in the second case **160**. In some embodiments, after electrical connection between the second connection tab **140** and the second case **160** is established, the electrode assembly **120** may be accommodated in the second case **160**.

[0096] The first case **150** may move downward (e.g., a direction towards the electrode assembly **120** and/or second case **160**) while surrounding the upper side of the electrode assembly **120** and may press (e.g., compress) the insulating gasket **170**. The first case **150** may be located outside the outer gasket **174** in an interference fit manner while compressing the outer gasket **174** of the insulating gasket **170**.

[0097] Electrical connection between the first connection tab **130** and the first case **150** may be established in a state in which the electrode assembly **120** is accommodated in the first case **150**. In other embodiments, after electrical connection between the first connection tab **130** and the first case **150** is achieved, the electrode assembly **120** may be accommodated in the first case **150**. Because the first case **150** and the second case **160** are mounted in an interference fit manner with the insulating gasket **170** interposed therebetween, the occurrence of short circuit due to contact therebetween may be prevented.

[0098] The insulating gasket **170** may be fixed to the end portion of the second case **160** through

interference fit, an adhesive, a fastening member, thermal bonding, mechanical coupling, chemical coupling, and/or the like.

[0099] The connection gasket **176** may be variously modified, so long as the connection gasket **176** interconnects the upper side of the inner gasket **172** and the upper side of the outer gasket **174**. The connection gasket **176** according to some embodiments of the present disclosure may include an inner chamber **177**, which is a space for impact absorption. The inner chamber **177** may define a single space or a plurality of partitioned spaces inside the connection gasket **176**, and may be mounted between the second case **160** and the first case **150**. Because impact transmitted to the first case **150** is absorbed by the inner chamber **177** formed in the insulating gasket **170**, the durability of the secondary battery **100** may be improved.

[0100] In the secondary battery **100** according to some embodiments of the present disclosure having the above configuration, external terminals are eliminated and/or significantly reduced and the first case **150** and the second case **160** have different polarities, and accordingly, the number of parts and production costs may be reduced. The first case **150** and the second case **160** may be directly connected to external terminals, and thus the freedom of design may be increased.

[0101] FIG. **8** is a perspective view showing an exemplary secondary battery **101** according to some embodiments of the present disclosure, and FIG. **9** is an exploded perspective view of the exemplary secondary battery **101** according to the present disclosure. As shown in FIGS. **8** and **9**, the secondary battery **101** according to another embodiment of the present disclosure may include an electrode assembly **120**, a first connection tab **130**, a second connection tab **140**, a first case **250**, a second case **260**, and an insulating gasket **270**. The first case **250**, the second case **260**, and the insulating gasket **270** of the secondary battery **101** according to the other embodiment of the present disclosure may be modified to have a rectangular shape, but the functions thereof are identical to the functions of those of the secondary battery **100** according to the above-described embodiment of the present disclosure. Thus, detailed description thereof will be omitted, and the same components are denoted by the same reference numerals.

[0102] The electrode assembly **120** may be accommodated together with an electrolyte in a space defined by the first case **250** and the second case **260**. The electrode assembly **120** may include a first electrode plate **121**, a second electrode plate **122**, and a separator **123** interposed between the first electrode plate **121** and the second electrode plate **122**. The components of the electrode assembly **120** may be stacked in a plate shape. The electrode assembly **120** may be configured such that the first electrode plate **121**, the separator **123**, and the second electrode plate **122** are stacked in that order. In some embodiments, the electrode assembly **120** may be configured such that the first electrode plate **121**, the second electrode plate **122**, and the separator **123**, each of which has a substantially rectangular plate shape, are stacked.

[0103] The first case **250** may be a rectangular cover having an open lower side to accommodate the electrode assembly **120**. The second case **260** may be a rectangular container having an open upper side to accommodate the electrode assembly **120**.

[0104] The first case **250** according to the other embodiment of the present disclosure may include a first case body **252** and a first extension member **254**. The first case body **252** may be a rectangular panel, to which the first connection tab **130** connected to the first electrode plate **121** may be electrically connected. The first extension member **254** may extend downward from the edge of the first case body **252** (e.g., away from the edge of the first case body **252** and/or towards the electrode assembly **120**) and may be located outside an outer gasket **274**. The first case **250** and the second case **260** may have shapes corresponding to each other.

[0105] The second case **260** according to the other embodiment of the present disclosure may include a second case body **262** and a second extension member **264**. The second case body **262** may be a rectangular panel, to which the second connection tab **140** connected to the second electrode plate **122** may be electrically connected. The second extension member **264** may extend upward from the edge of the second case body **262** and may be located outside the outer gasket

274.

[0106] The insulating gasket **270** may be formed in a rectangular frame shape corresponding to the shape of the first case **250**. The insulating gasket **270** may include an inner gasket **272** abutting the inner surface of the second case **260**, an outer gasket **274** abutting the outer surface of the second case **260**, and a connection gasket **276** interconnecting the inner gasket **272** and the outer gasket **274**. The inner gasket **272**, the outer gasket **274**, and the connection gasket **276** may be formed in a rectangular shape.

[0107] The above-described embodiments of the present disclosure may be variously modified, so long as electrical connection between the first case **150** or **250** and the second case **160** or **260** is blocked by the insulating gasket **170** or **270**, the first connection tab **130** is directly connected to the first case **150** or **250**, and the second connection tab **140** is directly connected to the second case **160** or **260**.

[0108] The batteries according to the above-described embodiments may be used to manufacture a battery pack.

[0109] FIGS. **10A** and **10B** are perspective views showing a battery pack including the exemplary secondary battery **100** or **101** according to the present disclosure. Referring to FIGS. **10A** and **10B**, the battery pack **300** may include a plurality of battery modules **200** and a housing **310** to accommodate the plurality of battery modules **200**. For example, the housing **310** may comprise a first and a second housing **311**, **312** that are coupled in facing directions with the plurality of battery modules **200** interposed between them. The plurality of battery modules **210** can be electrically connected to each other using a bus bar **251**, and the plurality of battery modules **200** can be electrically connected in series/parallel or a mixed series-parallel manner to obtain the required electrical output. In the drawings, for the sake of convenience, components such as bus bars, cooling units, and external terminals for the electrical connection of battery cells are omitted. In some embodiments, the battery pack **300** can be mounted on a vehicle. The vehicle may be, for example, an electric vehicle, a hybrid vehicle, or a plug-in hybrid vehicle. The vehicle can include both four-wheel and two-wheel vehicles.

[0110] FIGS. **11A** and **11B** are, respectively, a perspective view and a side view showing vehicles **400** and **500** including the exemplary battery pack **300** according to the present disclosure.

[0111] In FIG. **11A**, the battery pack **300** may include a battery pack cover **311**, which is part of the vehicle underbody **410** and may correspond to the first housing, and a pack frame **312**, which is placed beneath the vehicle underbody **410** and may correspond to the second housing. The battery pack cover **311** and pack frame **312** may be structurally integrated with the vehicle floor **420**. The vehicle underbody **410** separates the interior and exterior of the vehicle, and the pack frame **312** may be positioned outside the vehicle.

[0112] As shown in FIG. **11B**, the vehicle **500** can be assembled with additional components such as a hood **510** at the front of the vehicle body **400** and fenders **520** located at the front and rear of the vehicle. The vehicle **500** includes the battery pack **300** comprising the battery pack cover **311** and the pack frame **312**, and the battery pack **300** can be coupled to the vehicle body part **400**.

[0113] As is apparent from the above description, according to the present disclosure, because external terminals are eliminated and a first case and a second case have different polarities, the number of parts and production costs may be reduced.

[0114] According to the present disclosure, the first case and the second case may be connected to external terminals, and thus the freedom of design may be increased.

[0115] However, the effects achievable through the present invention are not limited to those described above, and other technical effects not mentioned can be clearly understood by those skilled in the art from the description of the invention provided below.

[0116] Although a few embodiments of the present disclosure have been shown and described, it would be appreciated by those skilled in the art that various changes and modifications may be

made in this embodiment without departing from the principles and spirit of the disclosure, the scope of which is defined in the appended claims and their equivalents.

Claims

1. A secondary battery comprising: an electrode assembly comprising a first electrode plate and a second electrode plate; a first case mounted in a shape surrounding at least one surface of the electrode assembly and electrically connected to the first electrode plate; a second case mounted in a shape surrounding at least an opposite surface of the electrode assembly and electrically connected to the second electrode plate; and an insulating gasket located between the first case and the second case, the insulating gasket comprising an insulative material, wherein electrical connection between the first case and the second case is blocked.
2. The secondary battery as claimed in claim 1, further comprising: a first connection tab connected to the first electrode plate and extending above the electrode assembly and/or in a direction towards the first case, such that the first connection tab is electrically connected to the first case; and a second connection tab connected to the second electrode plate and extending below the electrode assembly and/or in a direction towards the second case such that the second connection tab is electrically connected to the second case.
3. The secondary battery as claimed in claim 2, wherein the first connection tab is connected to one side of the first electrode plate in a longitudinal direction of the first electrode plate, and wherein the second connection tab is connected to one side of the second electrode plate in a longitudinal direction of the second electrode plate.
4. The secondary battery as claimed in claim 2, wherein the first connection tab is welded to the first case, and wherein the second connection tab is welded to the second case.
5. The secondary battery as claimed in claim 1, wherein the second case is disposed on the first case, and a first end portion of the second case is located inside the first case.
6. The secondary battery as claimed in claim 5, wherein the insulating gasket is mounted in a shape surrounding the first end portion of the second case to block electrical connection between the first case and the second case.
7. The secondary battery as claimed in claim 5, wherein the insulating gasket comprises: an inner gasket abutting an inner surface of the second case; an outer gasket abutting an outer surface of the second case; and a connection gasket interconnecting the inner gasket and the outer gasket.
8. The secondary battery as claimed in claim 7, wherein a length of the outer gasket in a vertical direction is greater than a length of the inner gasket in the vertical direction.
9. The secondary battery as claimed in claim 7, wherein a length of the second case in a vertical direction is greater than a length of the outer gasket in the vertical direction.
10. The secondary battery as claimed in claim 9, wherein the length of the outer gasket in the vertical direction is greater than a length of the first case in the vertical direction.
11. A secondary battery comprising: an electrode assembly comprising a first electrode plate and a second electrode plate; a first case configured to cover a first side of the electrode assembly and electrically connected to the first electrode plate; a second case electrically connected to the second electrode plate and configured to accommodate the electrode assembly, the second case comprising a side covered by the first case; and an insulating gasket located between the first case and the second case, the insulating gasket comprising an insulative material.
12. The secondary battery as claimed in claim 11, wherein the first case and the second case are formed as conductors and have different polarities.
13. The secondary battery as claimed in claim 11, wherein the insulating gasket comprises: an inner gasket abutting an inner surface of the second case; an outer gasket abutting an outer surface of the second case; and a connection gasket interconnecting the inner gasket and the outer gasket.
14. The secondary battery as claimed in claim 13, wherein the first case comprises: a first case

body formed to allow a first connection tab connected to the first electrode plate to be electrically connected thereto and to cover the first side of the electrode assembly; and a first extension member extending away from an edge of the first case body to be located outside the outer gasket.

15. The secondary battery as claimed in claim 14, wherein the second case comprises: a second case body formed to allow a second connection tab connected to the second electrode plate to be electrically connected thereto and to cover a second side of the electrode assembly; and a second extension member extending away from an edge of the second case body to be located between the outer gasket and the inner gasket.

16. The secondary battery as claimed in claim 15, wherein a length of the second extension member in a vertical direction is greater than a length of the outer gasket in the vertical direction.

17. The secondary battery as claimed in claim 15, wherein a length of the outer gasket in a vertical direction is greater than a length of the first extension member in the vertical direction.

18. A method of manufacturing a secondary battery, the method comprising: providing an insulating gasket located between a first case and a second case such that electrical connection between the first case and the second case is blocked, wherein: the first case mounted in a shape surrounding at least one surface of an electrode assembly, the electrode assembly comprising a first electrode plate and a second electrode plate, and electrically connected to the first electrode plate; the second case mounted in a shape surrounding at least an opposite surface of the electrode assembly and electrically connected to the second electrode plate; and the insulating gasket comprises an insulative material.

19. The method of claim 18, wherein the insulating gasket is mounted in a shape surrounding a first end portion of the second case to block electrical connection between the first case and the second case.

20. The method of claim 18, wherein a first connection tab, the first connection tab connected to the first electrode plate and extending above the electrode assembly and/or in a direction towards the first case, electrically connects the first case to the first electrode plate; and a second connection tab, the second connection tab connected to the second electrode plate and extending below the electrode assembly and/or in a direction towards the second case, electrically connects the second case to the second electrode plate.
